01 PLUG AND REAR PORT — POINTS, EDGES, MEASUREMENTS

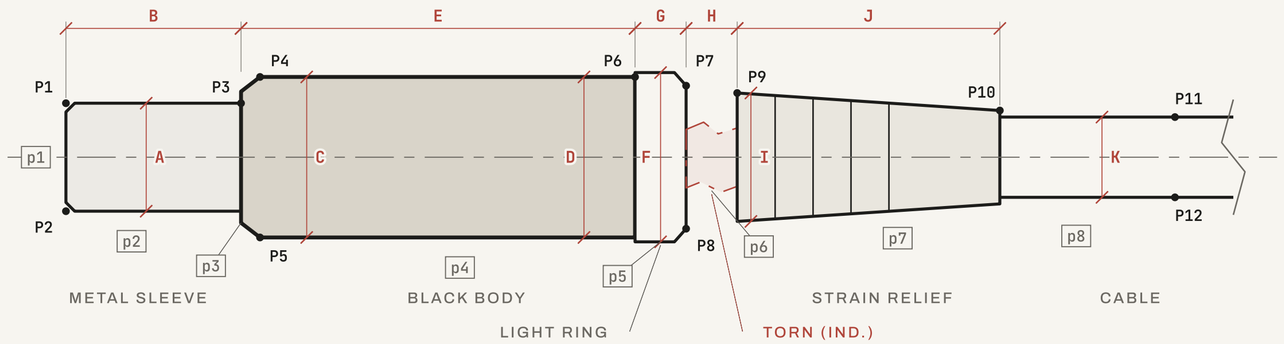
NOT TO SCALE · ALL DIMS IN mm · DRAWN FROM PHOTOS, NOTHING MEASURED YET

LAPTOP CHARGER BRACE

ALIENWARE m15 R7 · BARREL PLUG

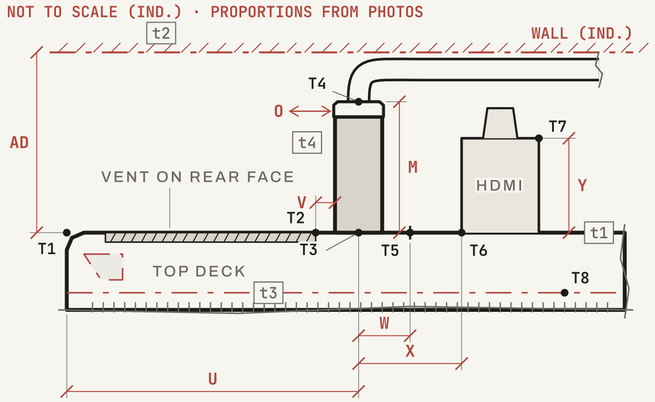
1 PLUG, SIDE ON

PLUG LYING ON THE DESK · METAL TIP ON THE LEFT · CABLE TO THE RIGHT
NOT TO SCALE (IND.) · PROPORTIONS FROM PHOTOS



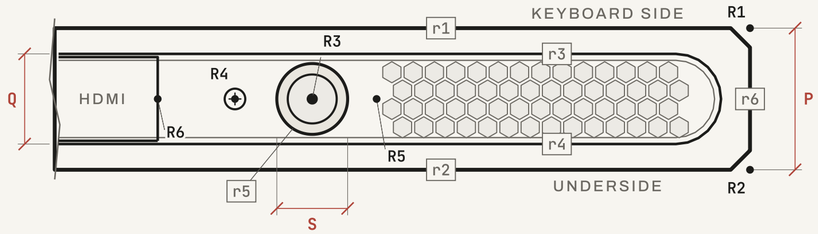
2 LAPTOP FROM ABOVE

KEYBOARD TOWARDS YOU · REAR EDGE AT THE TOP
VENT CORNER ON THE LEFT · PLUG IN
NOT TO SCALE (IND.) · PROPORTIONS FROM PHOTOS



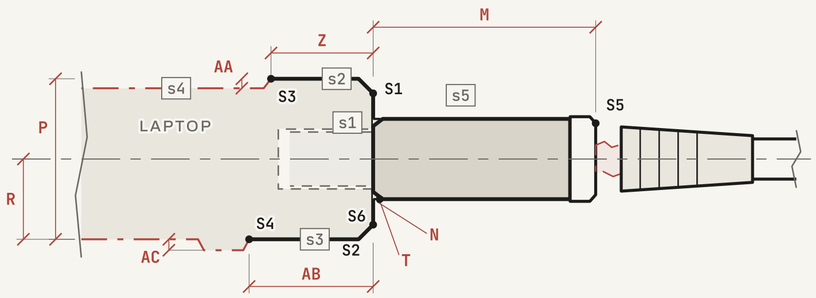
3 LAPTOP REAR FACE

REAR FACE TOWARDS YOU · KEYBOARD SIDE UP
MIRRORED FROM VIEW 2: VENT CORNER ON THE RIGHT
NOT TO SCALE (IND.) · PROPORTIONS FROM PHOTOS



4 SECTION THROUGH THE JACK

LOOKING FROM THE VENT CORNER · KEYBOARD LEFT · REAR FACE RIGHT · PLUG IN
NOT TO SCALE (IND.) · PROPORTIONS FROM PHOTOS



READING THIS SHEET

P1	POINT. The capital letter is the view: P plug, T top, R rear, S section.
p1	EDGE or surface. Lower-case, in a box. Same view letters.
A	MEASUREMENT. Red on the drawing until it is measured.
- - -	HIDDEN behind another part. A red broken line is drawn from a guess.
x y z	x along the rear from the vent corner · y up from the underside · z out of the rear face.

OPEN UNRESOLVED ON THIS SHEET

ALL	Nothing is measured. Check the drawing first, then measure A to AE by caliper. Scale bars follow.
A	7.4 is the nominal size of this type of barrel plug, not a reading. Confirm.
V4	Ridge, deck step, foot rib and hinge pocket are drawn from angled photos. Correct before measuring.
GRIP	How the brace holds the laptop is not decided: clip over the rear edge, the screw at T5, or adhesive.
TEAR	Bare cores at the tear. Insulate or replace the lead before it is braced: TBD.
AE	Where the laptop sits under the desk, and what is behind the plug there: TBD.

MEASUREMENT SCHEDULE

PLUG · VIEW 1

A	P1–P2	Metal sleeve outside diameter	7.4 (IND.)
B	P1→P3	Sleeve length, tip to the body	TBD
C	P4–P5	Body diameter at the sleeve end	TBD
D	p4 at P6	Body diameter at the ring end	TBD
E	P3→P6	Body length, sleeve to light ring	TBD
F	p5	Light ring diameter	TBD
G	P6→P7	Light ring length	TBD
H	P7→P9	Torn gap, cable pulled straight	TBD
I	at P9	Strain relief diameter, thick end	TBD
J	P9→P10	Strain relief length	TBD
K	P11–P12	Cable diameter	TBD

PLUG IN THE LAPTOP · VIEWS 2, 4

M	t1→T4	Rear face to the ring end, plug pushed home	TBD
N	s1→p3	Sleeve left showing, plug pushed home	TBD
O	at T4	Side-to-side play of the ring end	TBD

REAR FACE · VIEWS 2, 3

P	R1–R2	Laptop thickness at the rear	TBD
Q	r3–r4	Height of the recessed strip	TBD
S	r5	Jack collar outside diameter	TBD
T	S6→s1	Collar proud of (+) or sunk in (–) the strip	TBD
U	T1→T3	Vent corner to jack centre	TBD
V	T2→r5	Vent end to the jack collar	TBD
W	T3→T5	Jack centre to screw centre	TBD
X	T3→T6	Jack centre to the HDMI plug	TBD
Y	T6→T7	HDMI plug sticks out of the rear face	TBD

REAR PROFILE · VIEW 4

R	s3→axis	Underside to the centre of the jack	TBD
Z	s1→S3	Rear face to the ridge on the top deck	TBD
AA	s2→s4	Step height at the ridge	TBD
AB	s1→S4	Rear face to the foot rib underneath	TBD
AC	at S4	Foot rib height	TBD

ROOM BEHIND THE LAPTOP

AD	t1→t2	Rear face to the wall, laptop on the desk	TBD
AE	—	Rear face to nearest obstruction, under the desk	TBD

KEYED NOTES

1. **THE CABLE HAS ALREADY TORN AT THE LIGHT RING.** The strain relief has pulled out of the head and the cores are bare. The brace has to carry the cable past the tear, not only hold the head.
2. **THE VENT STARTS RIGHT BESIDE THE JACK.** Nothing may cover it. Any grip on the laptop goes above, below or towards the screw.
3. **THE PLUG LEAVES STRAIGHT OUT OF THE REAR.** Against the wall the cable is forced over at the head. The brace should turn it through a set radius instead.
4. **THE LAPTOP HAS NO LID.** The hinge pockets are open. A printed head is a later print on this same rear edge; the brace stays clear of it.

3D PRINTS · LAPTOP CHARGER BRACE

01 PLUG AND REAR PORT BLUEPRINT

NOT TO SCALE @ A2 · DIMS IN mm

REV A · 2026-10-04 · PALAPYE

STATUS: FIRST DRAWING — CHECK, THEN MEASURE